

Comparison Between Traditional and Alternated Resistance Exercises on Blood Pressure, Acute Neuromuscular Responses, and Rating of Perceived Exertion in Recreationally Resistance-Trained Men

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Abstract

Corrêa Neto, VG, Silva, DdN, Palma, A, de Oliveira, F, Vingren, JL, Marchetti, PH, da Silva Novaes, J, and Monteiro, ER. Comparison between traditional and alternated resistance exercises on blood pressure, acute neuromuscular responses, and rating of perceived exertion in recreationally resistance-trained men. *J Strength Cond Res* 38(5): e211–e218, 2024—The purpose of this study was to compare the acute effects of traditional and alternated resistance exercises on acute neuromuscular responses (maximum repetition performance, fatigue index, and volume load), rating of perceived exertion (RPE), and blood pressure (BP) in resistance-trained men. Fifteen recreationally resistance-trained men (age: 26.40 ± 4.15 years; height: 173 ± 5 cm, and total body mass: 78.12 ± 13.06 kg) were recruited and performed all 3 experimental conditions in a randomized order: (a) control (CON), (b) traditional (TRT), and (c) alternated (ART). Both conditions (TRT and ART) consisted of 5 sets of bilateral bench press, articulated bench press, back squat, and Smith back squat exercises at 80% 1RM until concentric muscular failure. The total number of repetitions performed across sets in the bench press followed a similar pattern for TRT and ART, with significant reductions between sets 3, 4, and 5 compared with set 1 ($p < 0.05$). There was a significant difference for set 4 between conditions with a lower number of repetitions performed in the TRT. The volume load was significantly higher for ART when compared with TRT. TRT showed significant reductions in BP after 10-, 40-, and 60-minute postexercise and when compared with CON after 40- and 60-minute postexercise. However, the effect size illustrated large reductions in systolic BP during recovery in both methods. Thus, it is concluded that both methods reduced postexercise BP.

Key Words: autonomic response, fatigue, performance, resistance training, strength

Introduction

Resistance training (RT) is an effective exercise modality for both improvements in muscle performance (39) and health (13). However, the adequate manipulation of the acute RT program variables including rest time between sets and exercises, intensity, and exercise order is fundamental to achieving the planned outcomes or goals (2). Specifically, the order of exercises can be an important strategy in the manipulation of acute physiological responses and thus ultimately in achieving the desired chronic adaptations. For example, resistance circuit-based training has

been considered an effective strategy to improve muscle performance and health (e.g., strength, body composition, and cardiorespiratory fitness (31)), and alternating upper and lower limbs in a non-circuit-format training session has been proposed as a time-efficient method, but there is a paucity of research regarding alternating whole-body multi-joint exercises (8).

Few studies have compared the effects of different RT protocols on cardiovascular responses to exercise in trained subjects. Previously, blood pressure (BP) response was analyzed after performing 2 different RT protocols (upper followed by lower limbs vs. alternating limbs) at intensities of 60 and 80% 1RM (3). The authors reported a reduction in systolic and diastolic BP after both protocols that remained until 15 minutes. Also, a significant reduction in systolic BP was observed at 30 minutes with intensities of

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80% of 1RM after both RT protocols (3). Similarly, 2 exercise orders were investigated but the authors did not find postexercise hypotension effect when manipulating the RT exercise order (16). It is noteworthy that both studies included in their respective RT protocols both multi- and single-joint exercises (3,16). It is well known that RT protocols that use greater muscle mass have a greater postexercise hypotension effect (17,30).

Manipulating training variables may exert an influence not only on health-related parameters but also on physical performance. Noninvasive markers have been used to assess performance response when manipulating RT variables, especially rating of perceived exertion (RPE) and fatigue index (FI) (38). It is well-defined that RPE is a useful tool when monitoring exercise intensity among different populations during RT protocols (5,24,25,36). Additionally, the FI can be an effective strategy to identify different fatigue levels caused by different RT protocols given its easy applicability (23). The possibility of improving the total training volume (TTV) when manipulating RT variables may also represent a performance marker because TTV seems to display an important role in muscle hypertrophy (21,34).

Therefore, the purpose of this study was to compare the acute effects of traditional and alternated resistance exercises on BP and neuromuscular responses (total number of repetitions, FI, TTV, and RPE) in recreationally resistance-trained men. The present study had 3 initial hypotheses. First, the traditional (TRT) condition will provide a greater decline in neuromuscular performance than the alternated (ART) condition. Second, the TRT condition would provide greater RPE than the ART condition. Third, the ART condition increases the magnitude of reductions in BP when compared with the TRT condition.

Methods

Experimental Approach to the Problem

The aim of this study was to compare the acute effects of traditional and alternated conditions on neuromuscular responses (maximum repetition performance, FI, and TTV), RPE, and BP in 15 resistance-trained men after 6 sessions. The first session was used to familiarize all subjects with all the experimental conditions. During sessions 2 and 3, the subjects underwent 1RM testing and retesting, respectively, for the bench press, articulated bench press, back squat, and Smith machine back squat exercises to determine the maximal external load to be used during both experimental conditions (TRT and ART). Then, subjects completed all 3 experimental conditions in a randomized order with a minimum of 48 hours interval: (a) control condition (CON), (b) TRT, and (c) ART. All procedures were performed at the same time of day (in the morning) to avoid any confounding circadian rhythm effect on BP. During the CON, after arriving for evaluation, subjects remained still for 10 minutes before the BP measurement. After that, BP measurements were taken in 10-minute cycles for 1 hour.

Subjects

Fifteen recreationally resistance-trained and nonhypertensive (2,40) men (Table 1) were recruited for the present study based on a previous sample size calculation study (23). All subjects had previous RT experience, including all exercises proposed in this study. All subjects were accustomed to performing at least 3 sessions per week with 50 minutes each and using external loads between 8 and 12 repetition maximum (RM) with rest intervals between 1 and

3 minutes between sets and exercises. Subjects had no functional limitations or medical conditions that could have compromised their health or confounded the study results. Before the study, all subjects were provided with verbal and written explanations of all study procedures and provided written informed consent to participate. Subjects then completed the Physical Activity Readiness Questionnaire and were only included in the study if they answered NO to all questions. The study was conducted in accordance with the norms of the Brazilian National Health Council, under resolution no. 466/2012, referring to scientific research on human subjects and the Helsinki Declaration, and was approved by the Augusto Motta University Centre ethics committee [# 5.007.390].

Procedures

A randomized (random entry in Latin Square Format) and counterbalanced within-subject experimental design was used to compare the effect of 2 RT conditions on BP, neuromuscular responses, and RPE. Subjects visited the laboratory 6 times during 16 days with at least 48 hours and no more than 72 hours between visits (Figure 1).

During the duration of their study, the subjects were instructed not to engage in any other RT session or strenuous physical activity. In addition, before data collection, the subjects were asked to confirm if they did not practice any other exercise session or strenuous physical activity. Subjects were asked to maintain their normal eating and sleeping habits, avoiding caffeine, ergogenic, and soothing consumption throughout the duration of the study. During TRT, all exercises comprising 1 specific body segment are executed before passing to any other body segment. Whereas the ART method alternates distinct body segments.

During the TRT and ART conditions, subjects performed all RT exercises until concentric muscular failure throughout 5 sets at 80% 1RM, with a self-suggested rest interval between sets and exercises (11,19), and the total number of repetitions in each set was recorded. The TRT condition was performed as follows: bench press, articulated bench press, back squat, and Smith machine back squat exercises, whereas the ART condition was performed as follows: the bench press, back squat, articulated bench press, and the Smith machine back squat. The bench press (Olympic Flat Bench, Model PG07; Technogym, Cesana, Italy) and the articulated bench press (Incline Chest Press, Pure Strength Line; Technogym) were performed in a lying (supine) position on the flat bench and a seating position on the bench, respectively, with the feet on a box, and the hands held the barbell with $1.5 \times$ the biacromial distance. Subjects were instructed to perform the concentric phase (horizontal shoulder adduction and elbow flexion) by moving the bar down until the arm was parallel to the ground and during the eccentric phase (horizontal shoulder abduction and elbow extension) returning the bar to the starting position. Back squat (Olympic Half Bench, Model PG07; Technogym) and Smith machine back

Table 1
Subject characteristics.

Age (y)	26.40 ± 4.15
Height (cm)	173 ± 5
Total mass (kg)	78.12 ± 13.06
Body mass index (kg/m ²)	26.17 ± 4.04
Resistance training experience (mo)	77.80 ± 50.62
Baseline systolic blood pressure (mm Hg)	133.73 ± 10.34
Baseline diastolic blood pressure (mm Hg)	75.47 ± 6.34

Table 2
Subject's load characteristics.*

Bench press 1RM (test) (kg)	84.27 ± 24.04
Bench press 1RM (retest) (kg)	85.20 ± 23.38
Relative muscle strength (×BW)	1.10 ± 0.24
Bench press 1RM (ICC)	0.995
Articulated bench press 1RM (test) (kg)	51.07 ± 13.16
Articulated bench press 1RM (retest) (kg)	53.60 ± 13.51
Relative muscle strength (×BW)	0.70 ± 0.15
Articulated bench press 1RM (ICC)	0.980
Back squat 1RM (test) (kg)	94.93 ± 34.79
Back squat 1RM (retest) (kg)	103.87 ± 31.49
Relative muscle strength (×BW)	1.33 ± 0.35
Back squat (ICC)	0.987
Smith back squat (test) (kg)	90.00 ± 33.17
Smith back squat (retest) (kg)	99.33 ± 33.64
Relative muscle strength (×BW)	1.27 ± 0.39
Smith back squat (ICC)	0.974

*1RM = 1 repetition maximum test; ICC = intraclass coefficient for 1RM test and retest.

squat (Multipower, Pure Strength Line; Technogym) were performed in a standing position with feet hip-width apart. Subjects were instructed to perform the eccentric phase (knee and hip flexion and ankle dorsiflexion) by moving the bar down until the thigh was parallel to the ground and during the concentric phase (knee and hip extension and ankle plantar flexion) returning the bar to the starting position.

Measurements and Calculations

One Repetition Maximum Testing. The subjects' 1RM was determined using a previously described method (4) for the bench press, articulated bench press, back squat, and Smith back squat exercises to define the external load used during both experimental protocols. Subjects performed a brief warm-up with 1 set

Table 4
Median (interquartile range) of the rating of perceived exertion values.*

	Set 1	Set 2	Set 3	Set 4	Set 5
Bench press					
ART	9 (8–9)	9 (9–10)§	10 (9–10)†§	10 (10–10)†§	10 (10–10)†
TRT	8 (8–9)	9 (8–9)	9 (9–10)	10 (9–10)†	10 (10–10)††
Articulated bench press					
ART	9 (9–10)§	10 (9–10)	10 (10–10)†	10 (10–10)†	10 (10–10)†
TRT	8 (8–9)	9 (9–10)	10 (10–10)†	10 (10–10)†	10 (10–10)†
Back squat					
ART	9 (8–9)	9 (9–10)§	10 (9–10)†	10 (10–10)†	10 (10–10)†
TRT	8 (8–9)	9 (9–9)	10 (9–10)†	10 (10–10)††	10 (10–10)††
Smith back squat					
ART	9 (8–10)	9 (9–10)	10 (10–10)	10 (10–10)	10 (10–10)†
TRT	9 (9–10)	10 (9–10)	10 (10–10)†	10 (10–10)†	10 (10–10)†

*ART = alternated resistance training; TRT = traditional resistance training.

†Significant difference to set 1, $p < 0.05$.

‡Significant difference to set 2, $p < 0.05$.

§Significant difference to TRT, $p < 0.05$.

of 10 repetitions at 40% of their estimated 1RM and 1 set of 3–5 repetitions at 60% of their estimated 1RM. The subjects were given a 1-minute rest between warm-up sets. Three minutes after the warm-up, the subjects performed the 1RM testing. They were allowed 3 attempts to obtain the 1RM load. Before testing, subjects were instructed on the proper form and execution of each exercise. The subjects performed a retesting 48 hours after the test to confirm the 1RM load. Consistent verbal encouragement was provided to the subjects. Subjects performed all exercises in a self-selected cadence. For the 1RM load, the interclass correlation coefficient (ICC) was calculated using the following equation: $ICC = [MS_b - MS_w] / [MS_b + \{k - 1\} \times MS_w]$, where MS_b = mean-square between, MS_w = mean-square within, and k = average group size. The same exercise

Table 3
Mean ± SD of the maximum number of repetitions per set for all experimental conditions.*

Condition	Set 1	Set 2	Set 2/set 1	Set 3	Set 3/set 1	Set 4	Set 4/set 1	Set 5	Set 5/set 1	FI (%)
Bench press										
ART	8.67 ± 1.84	7.60 ± 1.76	0.87 ± 0.95	7.00 ± 1.41†	0.80 ± 0.76	6.33 ± 1.54†	0.73 ± 0.83	5.13 ± 1.25†	0.59 ± 0.67	57.14 (50–70)
TRT	8.47 ± 1.55	7.80 ± 2.34	0.92 ± 1.50	7.00 ± 1.65†	0.82 ± 1.06	5.53 ± 1.64†	0.65 ± 1.05	4.60 ± 1.76†	0.54 ± 1.13	55.56 (42.86–66.67)
Articulated bench press										
ART	4.00 ± 1.00	4.07 ± 1.03	1.01 ± 1.03	3.27 ± 1.22†	0.81 ± 1.22	2.80 ± 0.86††§	0.70 ± 0.86	2.40 ± 1.24††	0.60 ± 1.24	50.00 (33.33–100)
TRT	4.00 ± 1.13	3.33 ± 1.40	0.83 ± 1.23	3.07 ± 1.39†	0.76 ± 1.23	2.47 ± 1.36†§	0.61 ± 1.20	2.67 ± 1.11†	0.66 ± 0.98	60.00 (50–75)
Back squat										
ART	12.00 (10–13)	12.00 (11–14)	1.05 ± 0.79	12.00 (10–13)	1.01 ± 0.84	11.00 (10–12)	0.94 ± 0.55	11.00 (9–12)†	0.94 ± 1.32	88.89 (75–100)
TRT	11.00 (9–12)	11.00 (10–12)	0.93 ± 0.40	10.00 (9–11)	0.84 ± 0.55	10.00 (9–11)	0.83 ± 0.57	9.00 (8–10)††	0.76 ± 0.38	80.00 (66.67–100)
Smith back squat										
ART	10.53 ± 2.23	10.07 ± 1.98	0.95 ± 0.88	9.47 ± 2.00	0.89 ± 0.89	8.80 ± 2.34	0.83 ± 1.04	9.60 ± 1.88	0.91 ± 0.84	100.00 (81.82–100)
TRT	8.40 ± 2.06	8.47 ± 1.30	1.00 ± 0.63	8.20 ± 2.01	0.97 ± 0.97	8.40 ± 2.44	1.00 ± 1.18	7.20 ± 1.74	0.85 ± 0.84	88.89 (62.5–100)

*ART = alternated resistance training; TRT = traditional resistance training; set 2/set 1 = repetitions in set 2 normalized to repetitions in set 1; set 3/set 1 = repetitions in set 3 normalized to repetitions in set 1; set 4/set 1 = repetitions in set 4 normalized to repetitions in set 1; set 5/set 1 = repetitions in set 5 normalized to repetitions in set 1; FI = fatigue index.

†Significant difference to set 1, $p < 0.05$.

‡Significant difference to set 2, $p < 0.05$.

§Significant difference to set 3, $p < 0.05$.

||Significant difference to TRT, $p < 0.05$.

Table 5
Median and Interquartile range of the time course for the blood pressure response (mm Hg) between protocols.*

	Baseline	Post-10	Post-20	Post-30	Post-40	Post-50	Post-60
Systolic blood pressure							
Control condition	134.00 ± 3.44	129.06 ± 3.73	134.13 ± 3.57	134.73 ± 2.86†‡	129.33 ± 3.60†	128.26 ± 3.89	135.60 ± 3.95†
ES		−1.37	0.03	0.23	−1.32	−1.56	0.43
		Large	No effect	Small	Large	Large	Small
Alternated resistance training	133.73 ± 2.67	126.86 ± 2.97	125.2 ± 2.95	126.66 ± 1.87	123.93 ± 2.85	124.66 ± 2.80	125.26 ± 2.50
ES		−2.43	−3.03	−3.11	−3.55	−3.31	−3.27
		Large	Large	Large	Large	Large	Large
Traditional resistance training	130.53 ± 3.05	124.13 ± 3.33†	125.73 ± 2.33	122.53 ± 3.65	120.06 ± 2.51†	123.46 ± 3.35	122.4 ± 3.65†
ES		−2.00	−1.78	−2.38	−3.76	−2.20	−2.42
		Large	Large	Large	Large	Large	Large
Diastolic blood pressure							
Control condition	77.93 ± 2.31	79.13 ± 3.68	78.06 ± 1.96	83.13 ± 3.43	76.26 ± 3.04	79.2 ± 3.14	77.06 ± 2.21
ES		0.40	0.06	1.81	−0.62	0.46	−0.38
		Small	No effect	Large	Medium	Small	Small
Alternated resistance training	75.46 ± 1.63	73.86 ± 2.30	75.06 ± 2.35	74.33 ± 2.22	74.4 ± 2.56	74.93 ± 3.03	77.2 ± 2.73
ES		−0.81	−0.20	−0.58	−0.50	−0.22	0.79
		Large	Small	Medium	Medium	Small	Medium
Traditional resistance training	76.8 ± 2.38	73.86 ± 2.51	74.66 ± 2.98	73.33 ± 2.58	72.20 ± 2.79	73.20 ± 2.19	73.26 ± 2.63
ES		−1.20	−0.79	−1.39	−1.77	−1.57	−1.41
		Large	Medium	Large	Large	Large	Large

*ES = effect size; IQR = interquartile range; Post-10 = 10 minutes postintervention; Post-20 = 20 minutes postintervention; Post-30 = 30 minutes postintervention; Post-40 = 40 minutes postintervention; Post-50 = 50 minutes postintervention; Post-60 = 60 minutes postintervention.

†Significant difference to baseline, $p < 0.05$.

‡Significant difference to alternated resistance training, $p < 0.05$.

apparatus was used for 1RM testing and during all experimental protocols (TRT and ART). The subject's load characteristics were provided in Table 2.

Neuromuscular Response. A bench press and back squat FI was determined from the total number of repetitions in the first (1st) and the last (5th) set and calculated using the equation proposed elsewhere (12), where a higher percentage value (%) indicates

a superior fatigue resistance: $FI = (5\text{th fifth set} \div 1\text{st set}) \times 100$. The TTV (TTV: $\sum[\text{total number of repetitions} \times \text{load for each exercise}]$) across each RT condition was calculated.

Rating of Perceived Exertion (OMNI-RES). After each RT condition, the subjects were asked to rate their overall effort using a scale ranging from 0 ("no effort") to 10 ("maximal effort and the most stressful exercise ever performed") (32).

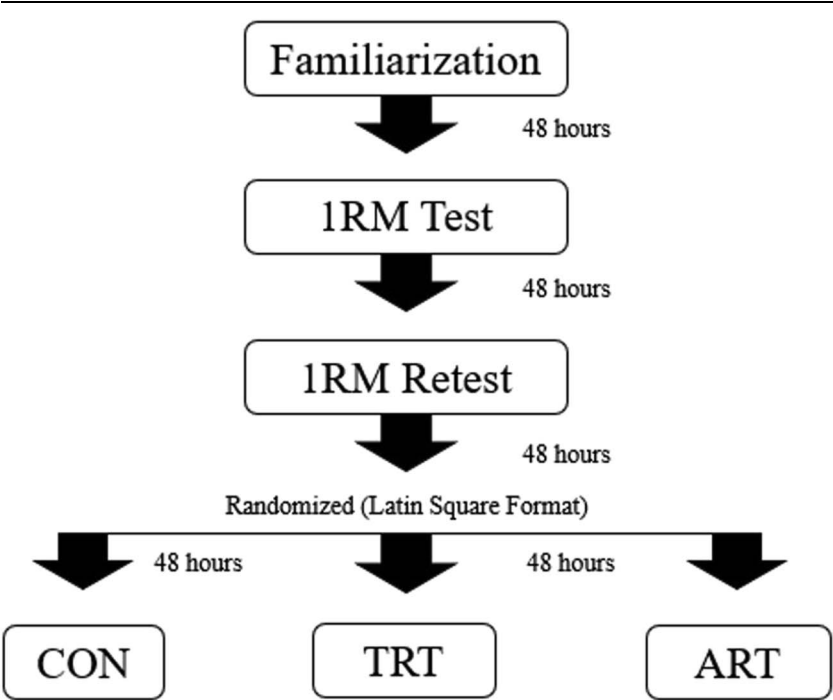


Figure 1. Study design. 1RM = one repetition maximum; CON = control condition without resistance exercises; TRT = tradition resistance training; ART = alternated resistance training.

Blood Pressure. Systolic and diastolic DBP were measured using an automatic oscillometer device (Omron Hem 7113, São Paulo, Brazil) (27). During each RT condition, BP was assessed after a 15-minute passive rest period after arrival at the laboratory (baseline: before any RT exercise) and at 10-, 20-, 30-, 50-, and 60-minute postexercise/rest condition (Post-10, Post-20, Post-30, Post-40, Post-50, and Post-60). Measurements were performed on the left arm consistent with the recommendations of the American Heart Association (1).

Statistical Analyses

The Shapiro-Wilk normality test was performed to assess data normality. When data normality was rejected, nonparametric tests were performed for statistical treatment. When the test was unable to reject data normality, symmetry, kurtosis, QQ plots, and graphical histogram analysis were conducted for confirmation. For BP, a 2-way ANOVA with repeated measures was used to compare the results within and between conditions followed by a post hoc Bonferroni test. Regarding TTV, pairwise comparisons were made through a Wilcoxon test. With regard to the total number of repetitions, data normality was rejected for the free-weight back squat and, therefore, a Friedman test with Bonferroni's correction was conducted. The maximum repetitions performance (for bench press, articulated bench press, and Smith back squat) had their normality accepted; therefore, a 2-way ANOVA with repeated measures was used to compare the results within and between conditions, followed by the Bonferroni post hoc test to determine pair-wise differences (26).

Concerning the FI, comparisons were performed using the Wilcoxon test. Finally, as a categorical variable, the RPE analysis was carried out using a Friedman test with adjusted pairwise comparisons within the conditions and the Wilcoxon test between conditions over the multiple sets for each exercise. Parametric data are presented as mean and SD. For nonparametric data, median and interquartile range values were presented as a measure of central tendency and descriptive statistics dispersion, which indicates the variability of results, respectively. The interquartile range was calculated based on dividing a data set into quartiles and subtracting the third quartile (upper limits) from the first quartile (lower limits).

Additionally, Cohen's *d* effect sizes were calculated using the formula $d = Md/Sd$, where *Md* is the mean difference and *Sd* is the SD of differences. This calculation differs slightly from traditional Cohen's *d* calculations, in that this formula better represents within-subject differences, whereas the traditional Cohen's *d* formula is better for between-subject comparisons. Cohen's *d* effect sizes were defined as small, medium, and large for 0.2, 0.5, and 0.8, respectively (9). An alpha level of 0.05 was used. All analyses were performed using SPSS version 21 (SPSS Inc., Armonk, NY).

Results

Descriptive Characteristics and Reproducibility of the 1RM Test

Table 1 provides mean values of total body mass, height, body mass index, age, baseline BP, and RT experience. Table 2 informs the mean values of relative muscle strength and intraclass correlation coefficient for the bench press, articulated bench press, back squat, and Smith back squat exercises. All exercises

presented intraclass correlation coefficient values above 0.9, classified as excellent reproducibility.

Maximum Number of Repetitions. The maximum number of repetitions (MNR) followed a similar pattern with significant reductions for sets 3, 4, and 5 compared with set 1 for both conditions (ART: set 3 vs. set 1, $p = 0.002$; set 4 vs. set 1, $p < 0.001$; set 5 vs. set 1, $p < 0.001$ and TRT: set 3 vs. set 1, $p = 0.006$; set 4 vs. set 1, $p < 0.001$; set 5 vs. set 1, $p < 0.001$; Table 3). There was also a significant difference between protocols for set 4 (MNR: ART > TRT, $p = 0.005$). For MNR, the ART condition showed a significant reduction in set 4 when compared with set 1 ($p = 0.002$) and set 5 when compared with set 1 ($p = 0.016$) for the articulated bench press exercise (Table 3). Significant reductions in the MNR were also observed between set 2 and set 3, set 4, and set 5 ($p < 0.001$) and between set 4 and set 3 ($p = 0.035$). In the TRT condition, there was a significant reduction in the MRP between set 1 and set 3 ($p = 0.002$), set 4 ($p < 0.001$) and set 5 ($p = 0.002$), and between set 4 and set 3 ($p = 0.004$). For the bench press, there was also a significant difference between RT conditions for set 2, with a lower MNR for the TRT condition ($p = 0.010$). For the back squat, the analysis within the ART indicated a significant reduction in the MRP between set 5 and set 2 ($p = 0.039$) and for TRT between set 5 and set 1 ($p = 0.022$) and set 2 ($p = 0.005$; Table 4). Significant differences were also observed in the comparison between RT conditions for the back squat exercise in sets 2 ($p = 0.012$), 3 ($p = 0.007$), 4 ($p = 0.038$), and 5 ($p = 0.020$), always with the lowest MRP for TRT. For the Smith back squat, there were only significant differences between RT protocols, and these differences indicated the lowest MRP for the TRT compared with ART for set 1 ($p = 0.001$), set 2 ($p = 0.004$), and set 5 ($p = 0.001$).

Fatigue Index. For FI, there was no significant difference between RT conditions for all RT exercises ($p > 0.05$). The MNR performance and FI values are shown in Table 3.

Total Training Volume. The TTV was significantly higher for the ART (2,676 [1,242–4,011]) when compared with the TRT condition (2,440 [1,095–3,864]; $p < 0.001$).

Rating of Perceived Exertion. For the RPE, there was a significant increase in the ART condition for the bench press exercise between set 1 and set 3 ($p = 0.003$), set 4 ($p < 0.001$), and set 5 ($p < 0.001$; Table 4). Similar responses were observed in the TRT condition, which showed significant increases compared between set 1 and set 4 ($p < 0.001$) and set 5 ($p < 0.001$), and between set 5 and set 2 ($p = 0.032$; Table 4). Significant differences were observed between RT conditions for set 2 ($p = 0.033$), set 3 ($p = 0.008$), and set 4 ($p = 0.25$), always with higher RPE for the ART (Table 4). For the articulated bench press exercise, the ART condition showed significant increases in RPE for set 3 ($p = 0.027$), set 4 ($p = 0.027$), and set 5 ($p = 0.027$), all compared with set 1 (Table 4). The TRT condition also presented significantly high RPE for set 3 ($p = 0.004$), set 4 ($p = 0.004$), and set 5 ($p = 0.001$) compared with set 1 (Table 4). Finally, a significant difference was observed between the RT conditions in set 1 ($p = 0.007$), with higher RPE for the ART condition (Table 4). For back squat, the ART protocol showed a significant increase in RPE in set 1 compared with set 3 ($p = 0.018$), set 4 ($p = 0.001$), and set 5 ($p < 0.001$; Table 4). In the TRT condition, significant increases in RPE also occurred between set 1 to set 3 ($p = 0.012$), set 4 ($p = 0.001$), and set 5 ($p < 0.001$; Table 4). In addition,

significant increase occurred between set 4 ($p = 0.022$) and set 5 ($p = 0.003$) in relation to set 2 (Table 4). When comparing RT conditions, RPE was significantly higher in set 2 ($p = 0.014$) during the ART condition (Table 4). In the Smith back squat, the ART condition showed a significant increase in RPE between set 5 and set 1 ($p = 0.005$), whereas in the TRT condition, significant increases in RPE were found between set 1 and set 3 ($p = 0.032$), set 4 ($p = 0.015$), and set 5 ($p = 0.007$; Table 4).

Blood Pressure. For systolic BP, the CON and ART conditions did not show any significant difference in postexercise ($p > 0.05$; Table 5). However, the TRT condition showed significant postexercise hypotension between the baseline and Post-10 ($p = 0.042$), Post-40 ($p = 0.001$), and Post-60 ($p = 0.025$; Table 5). Furthermore, the TRT condition showed significantly lower values than CON for Post-40 ($p = 0.026$) and Post-60 ($p = 0.004$; Table 5). Diastolic BP did not show significant difference changes during the follow-up time neither in the CON protocol nor in the RT conditions (ART and TRT; $p > 0.05$; Table 5).

Discussion

The purpose of this study was to compare the acute effects of 2 different RT conditions (traditional vs. alternated) on BP, neuromuscular responses (maximum repetition performance, FI, and TTV), and RPE in resistance-trained men. The main findings indicate that the TRT condition caused significant reductions in systolic BP during some postexercise moments. The present results confirmed the initial hypothesis regarding neuromuscular response indicating better neuromuscular performance (MRP: ART > TRT; VL: ART > TRT; FI: TRT = ART). However, the results do not confirm the hypothesis regarding the RPE, in which the results indicate higher values in ART. It is noteworthy that even in moments without significance, both RT conditions received an impact in the reduction of both systolic BP and diastolic BP over an hour after the RT session. Although reductions in BP levels may not have indicated statistical significance, the effect size portrays the impact of experimental conditions on BP reduction and these reductions may confer clinical benefit (28).

Limited studies aimed to analyze the BP responses after performing different RT conditions. An analysis of the BP responses after 2 RT methods (German volume training, GVT; and sarcoplasmic stimulating training, SST) was previously conducted (10). The authors observed BP reductions only when compared with the immediately post-RT and no difference when compared with the rest, which would mischaracterize the postexercise hypotension (10). Blood pressure reductions were found throughout different post-exercise times (e.g., Post-10, Post-20, Post-30, Post-40, Post-50, and Post-60) when compared with immediately postexercise time. Blood pressure reduction did not indicate significance in relation to the hypotensive responses, considering that the immediately postexercise recovery requires a few seconds and reflects an expected physiological response in healthy subjects (10,22). Thus, a BP reduction during recovery when compared with that time did not demonstrate either recovery to the peak exercise value or comparison with rest. Thus, the importance of considering the cardiovascular risk when thinking about different experimental designs in RT to promote postexercise hypotension is highlighted. The GVT and SST methods involve shorter rest intervals between sets, which may increase BP by a large magnitude during its performances (7). The present study used self-suggested rest intervals, which seems to be a positive strategy for hypertensive patients, without

compromising the safety of the RT performance because it ends up promoting more generous intervals, in the same way, that such a strategy does not harm the postexercise hypotension response (19).

Bentes et al. (3) previously analyzed the effects of ART and TRT on postexercise BP responses. The authors observed shorter hypotensive moments with no difference in magnitude effects regards to the method or intensity. The authors chose a sequence that restricted single-joint exercises when compared with multi-joint exercises. This approach may have influenced the results of Bentes et al. (3), which did not portray the effect size, thus making it impossible to discuss the results found in the present study. Within this context, de Freitas et al. (17) analyzed the BP responses after 3 RT protocols: a protocol involving only upper limb exercises, another lower limb protocol, and another a combination of upper and lower limbs. Regarding ST-volume equalization, only the protocol involving the combination of upper and lower limbs promoted significant BP reductions and confirmed the importance of activating larger muscle mass during an RT session. This hypothesis was previously raised by Polito et al. (30) when comparing the hypotensive effect of multiple sets performed in the biceps curl and the leg press and only observed significant postexercise BP reductions with a greater amount of muscle mass. Tomeleri et al. (37) compared the effect of different RT exercises orders on hypotensive responses and did not observe differences in either magnitude or duration of postexertion hypotension regardless of the order when the similarity of activated muscle mass was preserved. Halliwill (20) hypothesized, as a mechanism for post-exercise hypotension, the effect of local metabolites on vasodilation and the inhibition of alpha-adrenergic receptors, and consequently, these effects were more important than reductions in cardiac output that can be compensated by a reduced postexercise load and an increase in myocardial contractility. It is therefore pertinent to hypothesize a possible effect of the amount of muscle mass on the production of the local metabolites, which supports the results of the present study that used only multi-joint exercises and indicated large postexercise hypotension (according to the effect size) for both RT methods.

In the present study, the TRT condition provided a lower TTV (Table 4) because of a significant reduction in the maximal number of repetitions throughout sets when compared with the ART (Table 3). It may promote a significant reduction in the BP responses at some times (posttest) after the RT session (Table 5). Previous studies have indicated that a larger RT volume might be more effective in providing postexercise hypotension in young adults; however, these studies did not compare different RT methods but with a different number of sets. Rodriguez et al. (33) compared the effects of the TRT and the tri-set method on the BP responses after RT method and no statistically significant change was identified in the BP values during the postexercise recovery period. Considering that the same exercises were selected for both RT methods, the area of muscle mass activation was similar. Rodriguez et al. (33) cited that only upper limb exercises were used, and it may have compromised their results because of the smaller amount of muscle mass activated. Thus, when comparing different RT methods, the higher activated amount of muscle mass seems to be more relevant than the TTV, considering that the results of the present study did not follow the hegemonically expected pattern that the higher volume would lead to a greater postexercise hypotension response. However, even though with statistical significance only for systolic BP caused by the TRT, both conditions promote BP decreases that can have a clinical

impact on both systolic and diastolic BP, illustrated by the effect size. Polito et al. (30) and Bentes et al. (3) did not discuss the effect size of their results, limiting a richer discuss with the findings of the present investigation.

Although the FI (Table 3) did not show a significant difference in any of the exercises when comparing the 2 RT conditions (TRT and ART), the RPE (Table 5) was higher in some periods (posttest) for the AST when compared with the TRT. This fact may be related to the higher maximal number of repetitions performed in some sets in the ART condition (Table 3). The maximal number of repetitions seems to have a greater impact on RPE than the exercise order (35,36). Considering that the recommendations for the practice of physical activity, as a component of the nonpharmacological treatment of arterial hypertension, presuppose regularity (29) strategies that denote a lower RPE and, consequently, may facilitate the RT adherence (18).

There are a few limitations and delimitations to note in the interpretation of this study. First, interindividual differences in the time under tension (TRT and ART) were not tightly controlled. This can be considered as both a limitation and a strength of this design. Specifically, the lack of control reduces the internal validity of the results because the number and duration of each repetition may influence the outcome. Conversely, the freedom to choose the pace and duration of each repetition enhances the ecological validity and generalizability of the findings. Second, the seated posture for recovery was adopted in accordance with the American Heart Association (1,14,15). This posture may hinder venous return, overload cardiovascular function, and affect BP values because of mechanical obstruction of blood vessels by body mass and reduce the performance of the lower-extremity skeletal muscle venous pump postexercise. However, this position is recommended for the assessment of BP (1). Finally, it is noteworthy that attempts were made to minimize the impact of the circadian cycle on the posteffort BP response following the same period of the day for data collection (6).

According to the observed results, both methods are effective in reducing postexercise BP levels, and the inclusion of both ART and TRT in RT routines that aim to provide BP reductions in the succeeding moments seems to be an interesting strategy.

Practical Applications

Blood pressure responses should be taken into consideration after RT exercises. This study suggests that BP decreases after both TRT and ART protocols with 5 sets of the upper (bench press and articulated bench press) and lower (back squat and Smith back squat) multi-joint exercises. We found decreases in systolic BP by TRT and ART with a large and meaningful magnitude (effect size) in normotensive men. This may have an impact on the design and implementation, in both athletic and rehabilitation subjects, and may help to influence decision-making by practitioners who desire to elicit a post-exercise hypotension response in both normotensive and hypertensive participants. In addition, although investigating chronic changes in post-exercise hypotension as a response to these interventions was beyond the scope of this current investigation, there are ample data to suggest that RT can be a powerful tool for BP reduction. Practitioners who are looking to BP influence parameters for their clients may wish to explore incorporating RT in their plan of care.

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